



**SRI SHANMUGHA COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)**

**Approved by AICTE, Affiliated to Anna University,
Accredited by NAAC, NBA (ECE/CSE/MECH) and ISO 9001:2015
Certified
Pullipalayam, Sankari, Salem (Dt.)**

Program Outcomes (POs)

- PO1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

- PO9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

- PSO 1.** Ability to identify, design and apply domain knowledge and computational skills to evolve novel intelligent solutions for Artificial Intelligence and Data Science related processes in our ecosystem.
- PSO 2.** Ability to critique the role of Artificial Intelligence and Data Science in multi-disciplinary areas to transform thoughts into products through research and innovative career.
- PSO 3.** Ability to Apply adaptive machine learning algorithms, statistical models, tools and techniques to develop intelligent systems for solving problems from inter-disciplinary domains.

Program Educational Outcomes (PEOs)

- PEO 1.** Graduates will be able to demonstrate their technical skills and competency in various applications by providing creative and novel technological solutions.
- PEO 2.** Graduates will be able to ensure the effective contribution to the society through critical thinking, innovations and research with the broad spectrum of skills in frontline technologies.
- PEO 3.** Graduates will be equipped with multidisciplinary initiatives and visions towards the growth of society with respect to ethical and lifelong learning.